

STATISTICS

PART 1: Basic Concepts, Graphical Projection of Data, Frequency Distributions

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Rev:03.10.2023

Basic Concepts of Statistics

Statistics

Engineering, scientific, industrial, etc. problems can be solved through

Analytical Methods

(Exact or approximate solutions)

Numerical Methods

(Approximate solutions)

Statistical Methods or Artificial Intelligence Methods

(Estimated solutions)

Deterministic Approach

The problem is investigated under specific conditions.

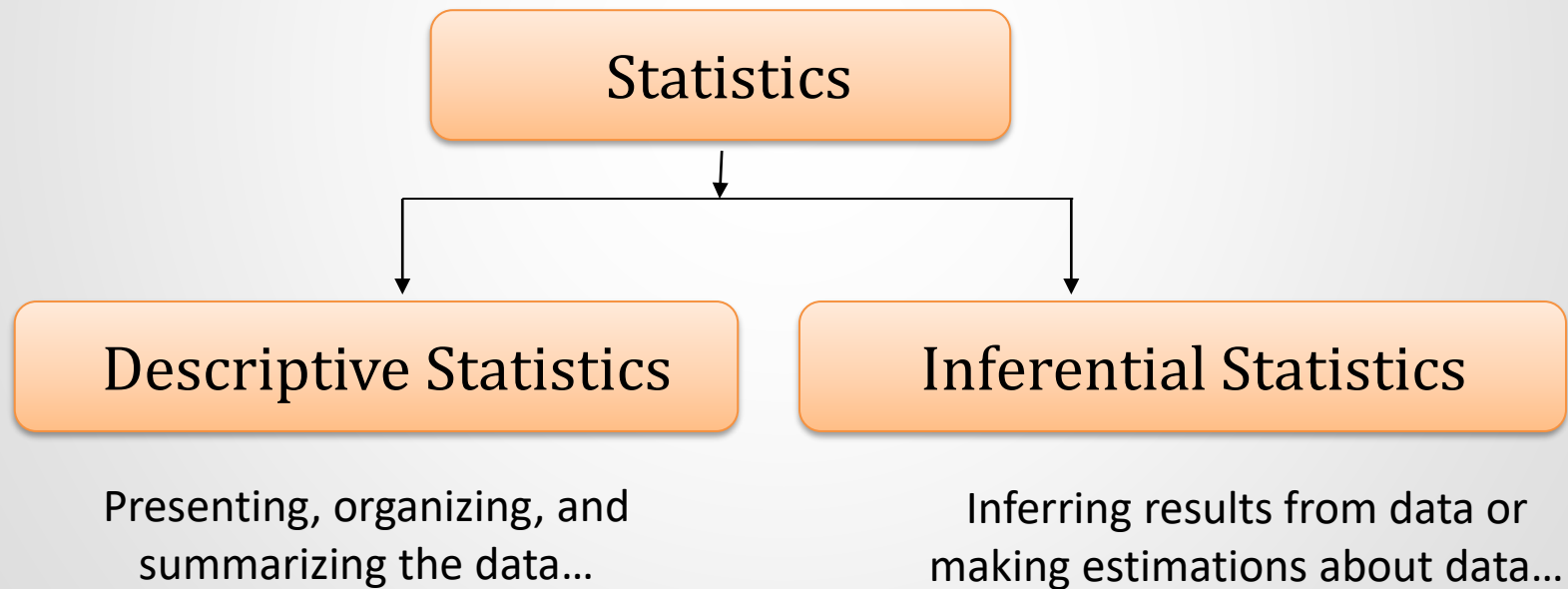
Stochastic Approach

The problem is investigated under probabilistic conditions by using collected data.

Basic Concepts of Statistics

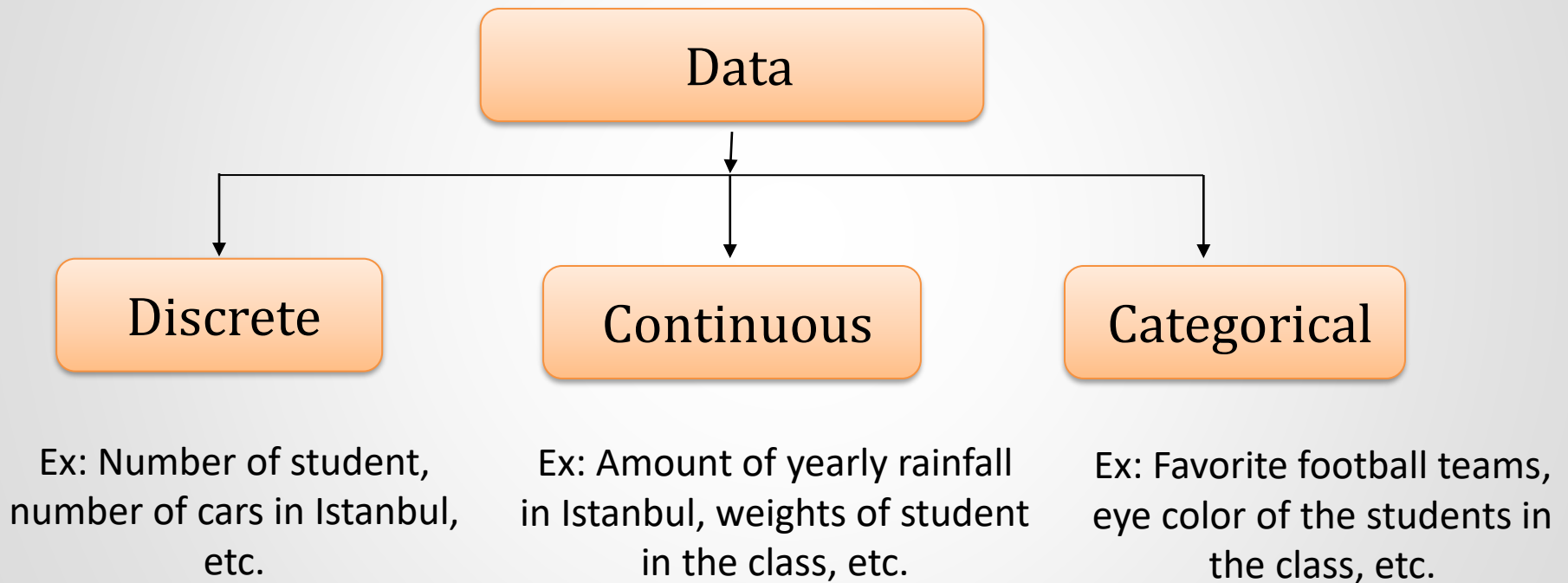
Statistics

Definition: **Statistics** is the science of collection, organization, interpretation, and estimation of data.



Basic Concepts of Statistics

Statistics



Graphical Projection of Data

Statistics

Consider the following illustrative example...

EXAMPLE 1 (PUZZLE EXAMPLE): In order to learn in how many minutes a newly designed puzzle can be solved, 50 children were observed when they solved this puzzle. The following table shows solving times (in minutes) for the puzzle:

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19

Graphical Projection of Data

Statistics

How can we describe the given data?

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19



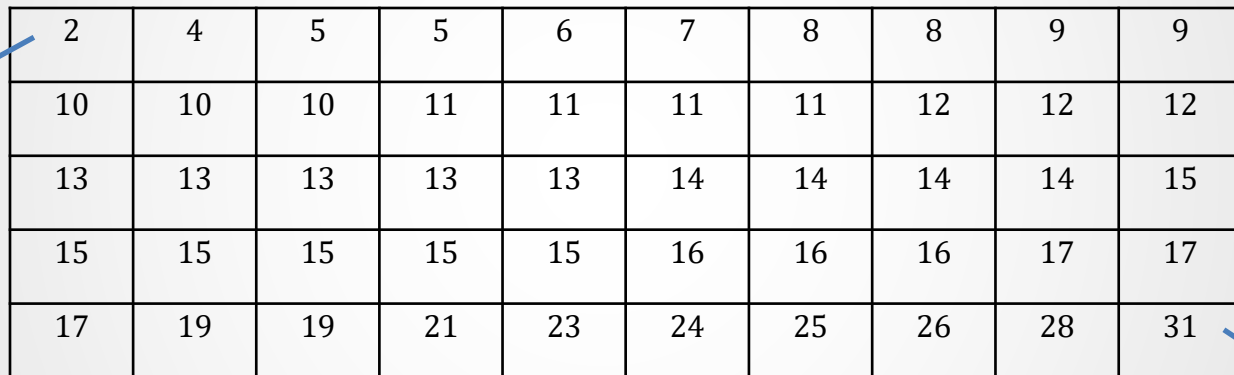
The simplest organization of the data is sorting them.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Graphical Projection of Data

Statistics

The Sorted Data



2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Min. value

Max. value

$$n = 50$$

$$\begin{aligned}\text{Range of the data} &= \text{Max. Value} - \text{Min value} \\ &= 31 - 2 \\ &= 29\end{aligned}$$

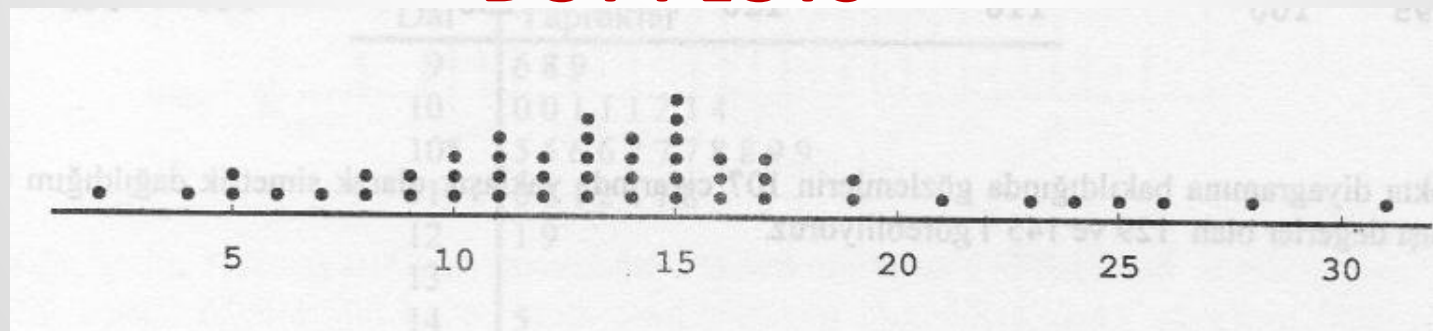
Graphical Projection of Data

Statistics

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

There are many different ways to illustrate data.

DOT PLOTS



Dot plots can be used to see the distribution of data, where data are concentrated or extreme values, etc.

Graphical Projection of Data

Statistics



2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

STEM-AND-LEAF PLOTS

Stem	Leaf
0	2 4 5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4 5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4 5 6 8
3	1

Since the given data set consists of one and two-digit numbers, here stem represents the tens-digit of the values while the leaf represents the ones-digit of the values.

Graphical Projection of Data

Statistics



2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

In order to increase more illustrative stem-and-leaf plots, we can increase the number of stems.

Stem	Leaf
0	2 4
0*	5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4
1*	5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4
2*	5 6 8
3	1

Graphical Projection of Data

Statistics



2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

FREQUENCY DISTRIBUTION

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

Another way of showing the frequency of values in a data set is using a **frequency distribution table**.

$$\sum_i f_i = n = 50$$

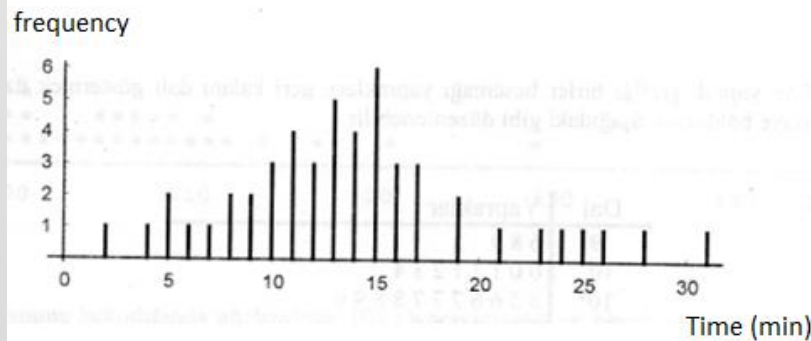
Graphical Projection of Data

Statistics

LINE PLOT

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

For the graphical display of frequency distribution tables, we often use **line plots**.



Graphical Projection of Data

Statistics



CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

When a data set contains a large number of discrete data or continuous data, it may not be appropriate to use frequency distribution. For such cases, we can use **classified frequency distribution**.

Steps for Creating Classified Frequency Distribution:

1. Compute the range of data.
2. Determine the number of classes (k). In practice, k can be chosen generally between 5-15. You may use the following formulas to specify it:

$$k = \sqrt{n}$$
$$k = 1 + 3.3 \log n$$
$$k = 5 \log n$$

Graphical Projection of Data

Statistics



CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

3. Divide the data into classes, which are sub-groups with equal intervals. To create these classes, first determine the class interval:

$$\text{Class interval} = \left[\frac{\text{Range of data}}{\text{Number of classes } (k)} \right]$$

4. Find the midpoints (x_i) of each class.
5. Compute the frequency (f_i) of each class.

Graphical Projection of Data

Statistics

Ex: Create a classified frequency distribution table for the puzzle example.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Range = 29

For the given data set, we choose k as 6. (Here we have not used the given formulas.)

$$\text{Class interval} = \left\lceil \frac{29}{6} \right\rceil = 5$$

Graphical Projection of Data

Statistics



2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2
		$\sum_{i=1}^k f_i = 50$

Attention: In this course, we will assume that the value on a lower bound belongs to that class, but the value on the upper bound does not belong.

For example, the value '7' belongs to the second class (7-12).

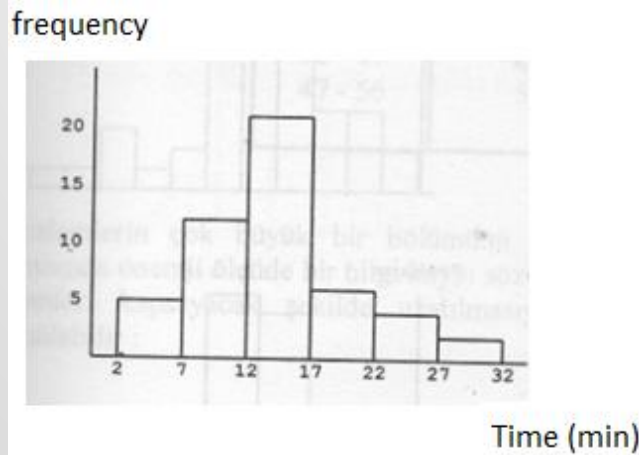
Graphical Projection of Data

Statistics

HISTOGRAMS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2

For the graphical display of classified frequency distribution tables, we use **histograms** instead of line plots.



Here, rectangles represent the classes or mid-points.

Graphical Projection of Data

Statistics

CLASSES WITH UNEQUAL INTERVALS

Assume that 4 new data are added to the data set: 28, 34, 41 and 55
In this case;

$$\text{Range} = 55 - 2 = 53$$

If we select k as 6 again,

$$\text{Class interval} = \left\lceil \frac{53}{6} \right\rceil = 9$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-11	6,5	13
11-20	15,5	30
20-29	24,5	7
29-38	33,5	2
38-47	42,5	1
47-56	51,5	1
		Sum : 54

As seen from the table of classified frequency distribution, most of the data are concentrated in the second class. Thus, we have a significant loss of information (distribution of data). Here, the problem occurs due to extreme values, i.e., 41 and 55.

Graphical Projection of Data

Statistics

CLASSES WITH UNEQUAL INTERVALS

To overcome this problem, we can choose class intervals as 5 again, then extend the interval of the last class such as given below:

Classes (C_i)	Frequency (f_i)
2-11	13
11-20	30
20-29	7
29-38	2
38-47	1
47-56	1

For classes with
equal interval

Classes (C_i)	Frequency (f_i)
2-7	5
7-12	12
12-17	21
17-22	6
22-27	4
27-57	6

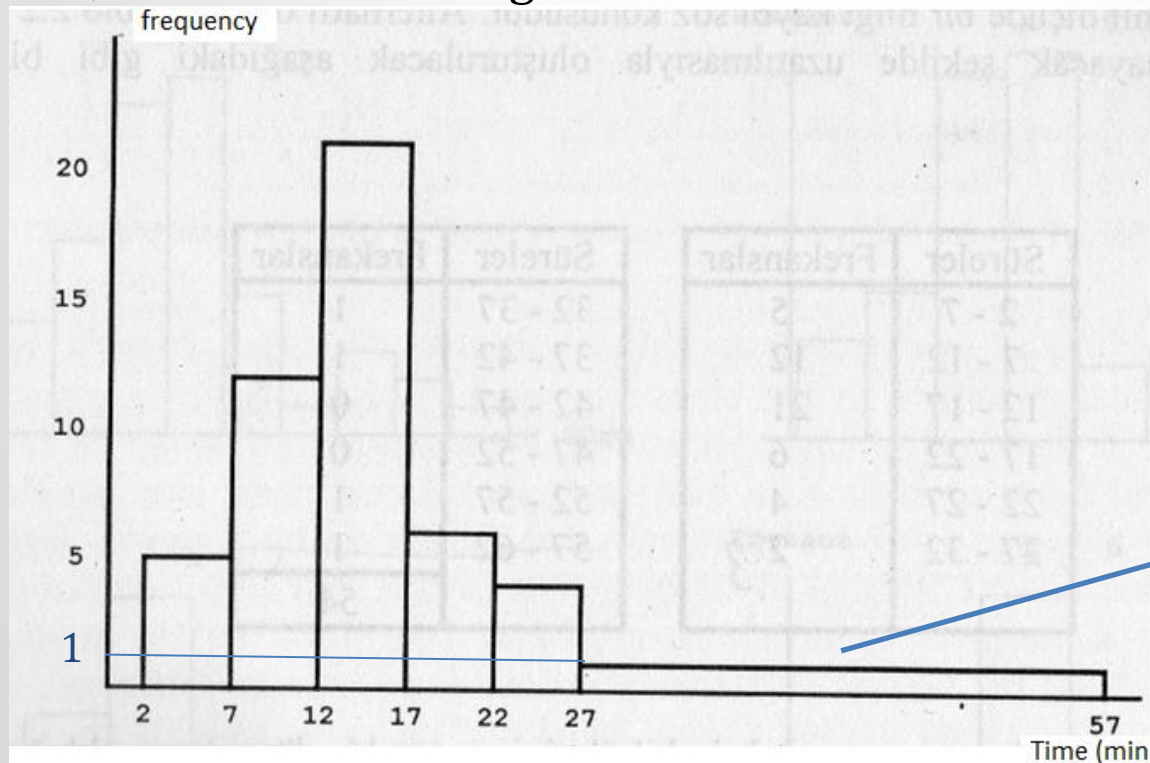
For classes with
unequal interval

Graphical Projection of Data

Statistics

HISTOGRAM FOR CLASSES WITH UNEQUAL INTERVALS

The area of each rectangle in a histogram represents the corresponding frequency if the class intervals (width of rectangles) are normalized with 1. Then, the sum of rectangle areas will be the number of data, i.e., n .



Why is the height of the rectangle 1 instead of 6?

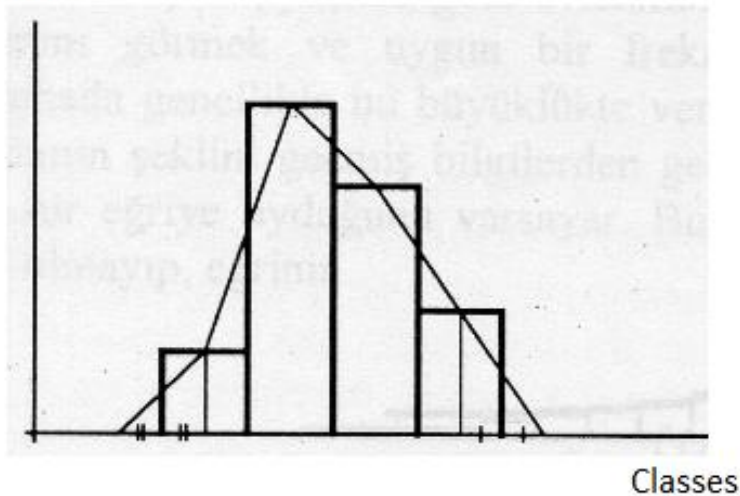
Graphical Projection of Data

Statistics

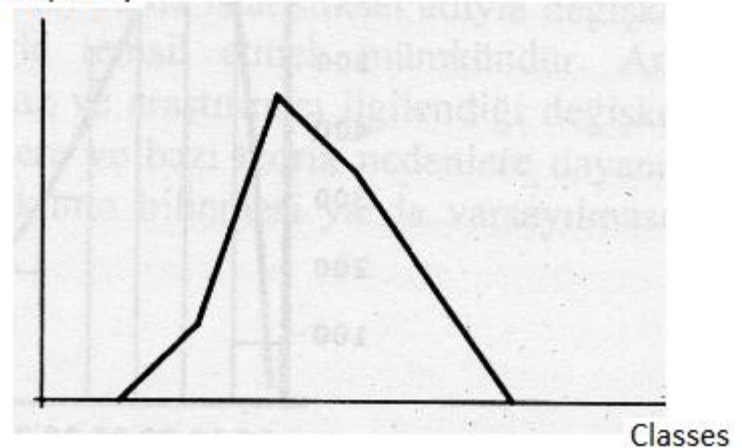
FREQUENCY POLYGON AND FREQUENCY CURVE

A frequency polygon consists of the lines between midpoints of rectangles in a histogram. The first and the last line intersect the horizontal axis. The area below the frequency polygon is equal to the total area of rectangles.

Frequency



Frequency



Frequency Polygon

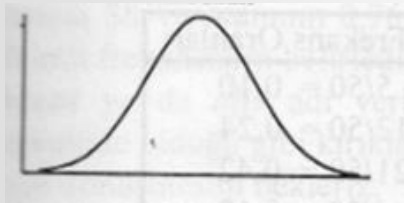
Graphical Projection of Data

Statistics

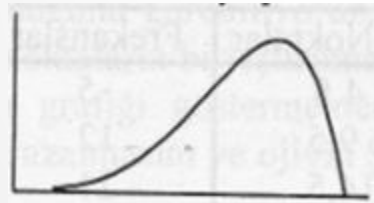
FREQUENCY POLYGON AND FREQUENCY CURVE

When the number of data (accordingly the number of classes) increases, the frequency polygon gets smoother. In this case, this polygon is called a **frequency curve**.

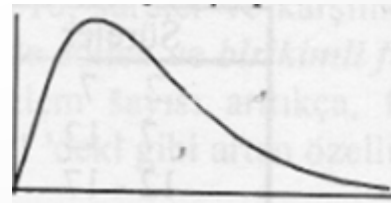
Some Basic Types of Frequency Curves



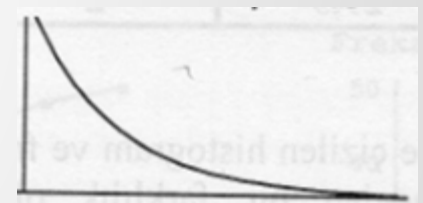
Symmetric



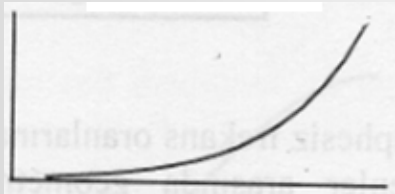
Skewed to left



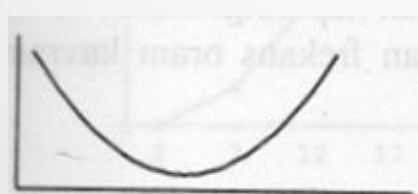
Skewed to right



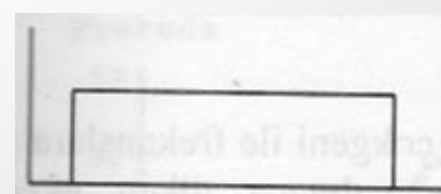
Inverse J-Shaped



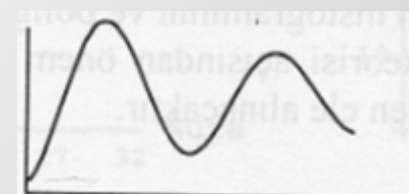
J-Shaped



U-Shaped



Rectangle



Two-mode

Graphical Projection of Data

Statistics

RELATIVE FREQUENCY DISTRIBUTION

The **relative frequency** of a class $i, i=1,2,...,k$, can be computed as:

$$\text{Relative Frequency } (f_i^R) = \frac{\text{Frequency } (f_i)}{\text{Number of data } (n)}$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)
2-7	4,5	5	5/50=0,10
7-12	9,5	12	12/50=0,24
12-17	14,5	21	21/50=0,42
17-22	19,5	6	6/50=0,12
22-27	24,5	4	4/50=0,08
27-32	29,5	2	2/50=0,04

$$\sum_i f_i^R = 1$$

Note: For relative frequency, the area of the histogram is 1 if the width of each rectangle is 1. Accordingly, the area under the frequency polygon or curve is 1.

Discuss the relationship between this fact and probability functions!!!

Graphical Projection of Data

Statistics

(INCREASING) CUMULATIVE FREQUENCY DISTRIBUTIONS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Increasing) Cumulative Frequency ($F_i \nearrow$)	Cumulative Relative Frequency ($F_i^R \nearrow$)
2-7	4,5	5	$5/50=0,10$	5	0,10
7-12	9,5	12	$12/50=0,24$	17	0,34
12-17	14,5	21	$21/50=0,42$	38	0,76
17-22	19,5	6	$6/50=0,12$	44	0,88
22-27	24,5	4	$4/50=0,08$	48	0,96
27-32	29,5	2	$2/50=0,04$	50	1,00

Graphical Projection of Data

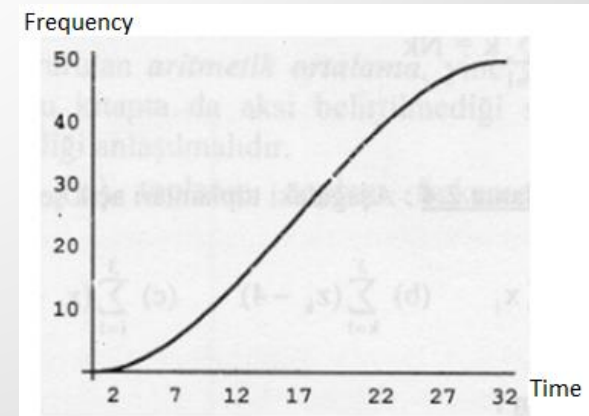
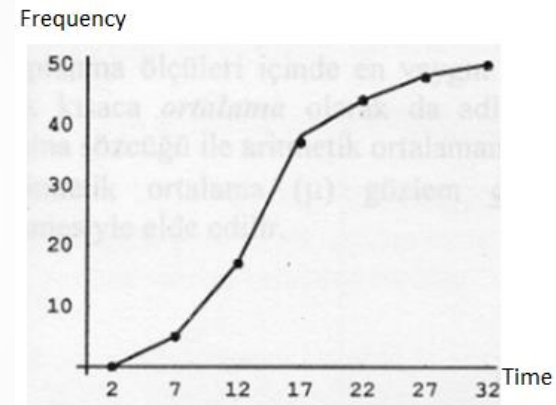
Statistics

CUMULATIVE FREQUENCY POLYGON AND CURVE

Classes (C_i)	(Increasing) Cumulative Frequency ($F_i \nearrow$)
2-7	5
7-12	17
12-17	38
17-22	44
22-27	48
27-32	50

Cumulative
Freq. Polygon

Cumulative
Freq. Curve



Graphical Projection of Data

Statistics

(DECREASING) CUMULATIVE FREQUENCY DISTRIBUTION

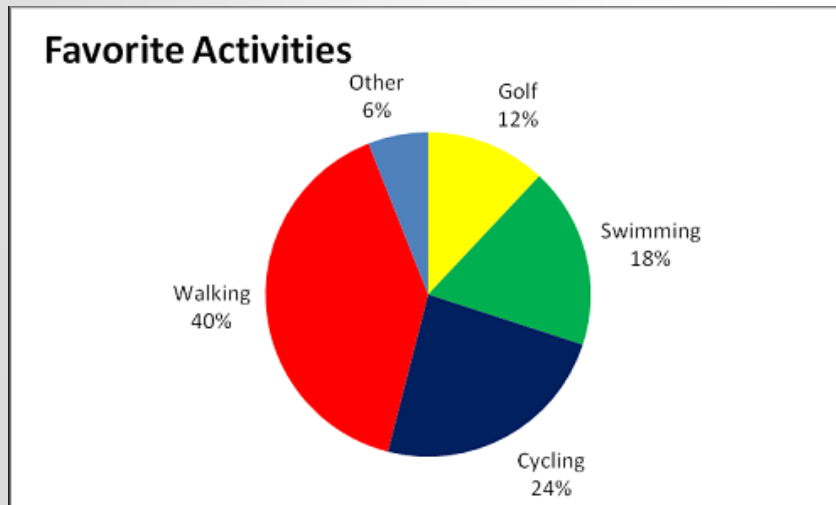
Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Decreasing) Cumulative Frequency ($F_i^{\searrow}$)	Cumulative Relative Frequency ($F_i^{R \searrow}$)
2-7	4,5	5	$5/50=0,10$	50	1,00
7-12	9,5	12	$12/50=0,24$	45	0,90
12-17	14,5	21	$21/50=0,42$	33	0,66
17-22	19,5	6	$6/50=0,12$	12	0,24
22-27	24,5	4	$4/50=0,08$	6	0,12
27-32	29,5	2	$2/50=0,04$	2	0,04

Graphical Projection of Data

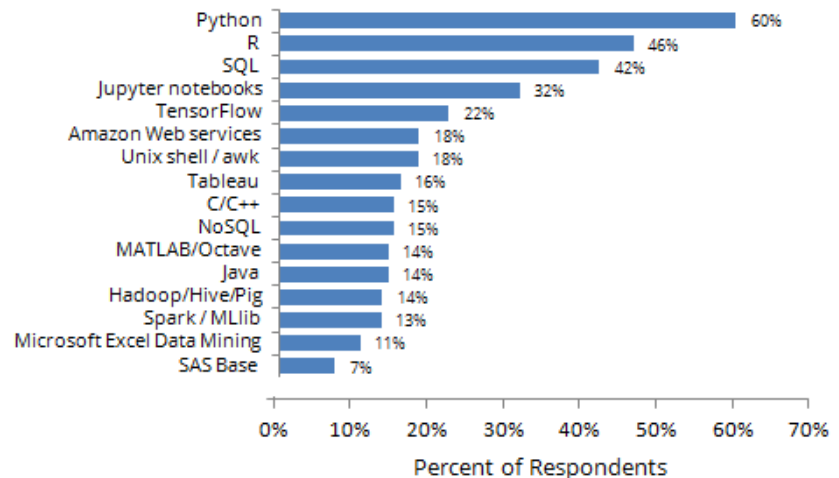
Statistics

For the projection of numerical data, we have used dot plots, line plots, histograms, frequency polygons, etc. When data are categorical, we can use many figures such as circular (pie) charts and bar charts.

Data science tools, technologies, languages used in 2017



A Circular Chart (1)



A Bar Chart (2)

(1) <https://www.open.edu/openlearncreate/mod/oucontent/view.php?id=569&printable=1>

(2) <https://businessoverbroadway.com/2018/01/02/most-used-data-science-tools-and-technologies-in-2017-and-what-to-expect-for-2018/>